HUANQI WU

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EDUCATION

University of Arizona — Tucson, AZ, United States
B.S. in Data Science | GPA 4.00/4.00 — Sept 2023 - May 2027
Capital University of Economics and Business — Beijing, China
B.S. in Statistics | GPA 3.78/4.5 (90.13/100) — Sept 2023 - May 2027
Relevant Coursework: Challenges in Biomedical Engineering, Bioinformatics and Functional Genomic Analysis, Bayesian Modeling & Inference, Introduction to Statistical Machine Learning, Theory of Probability, Multivariate Statistics, Sampling Theory
Academic Distinction: Dean's List (Spring 2025, Fall 2025, Spring 2026)

PROFESSIONAL EXPERIENCE

Genetron Health — Beijing, China
Bioinformatics Engineer — Jul - Sep 2026
• Reviewed BERT-based models for DNA-methylation representation learning and prediction, assessing their feasibility for liquid-biopsy early cancer detection.
• Curated large-scale methylation datasets from multiple public repositories; built end-to-end pipelines for sample filtering, metadata parsing, normalization, and model-ready preprocessing.
• Analyzed heterogeneous methylation cohorts to assess sequencing quality, batch effects, and systematic biases.
• Benchmarked deep-learning models using cancer-associated methylation markers and assessed feature-set robustness in distinguishing malignant from normal epigenomes.
• Documented experimental settings and workflows in technical reports; collaborated with wet-lab teams to align analyses with clinical assay requirements.

In-Capital — Beijing, China
Data Analyst — Nov 2024-Jan 2025
• Compiled comprehensive M&A screening reports for secondary-market healthcare targets, performed due-diligence research to evaluate feasibility of reverse-takeover (RTO) listing strategies for biotech and medical enterprises.
• Aggregated and structured multidimensional datasets covering corporate fundamentals, historical transaction records and healthcare capital-market dynamics to inform internal investment-review committee deliberations.
• Conducted comparative benchmarking across peer medical-industry players; sorted out valuation metrics, deal structures and regulatory constraints relevant to healthcare-sector back-door listing transactions.
• Synthesized market intelligence and analytical findings into investment memorandums, delivered data references to support fund partners' strategic investment decision-making.

RESEARCH EXPERIENCE

Real-World Evidence of Enteric-Release Budesonide in IgA Nephropathy
(Third Author, Manuscript under review) — *Data Analyst, Jun 2026*
• Analyzed 135 repeated observations from 38 patients via longitudinal mixed-effects models, evaluated longitudinal trajectory changes of proteinuria and renal-function indicators.
• Conducted stratified subgroup analyses to explore treatment-associated differences across patient clinical-phenotype subgroups.
• Processed and cleaned real-world clinical observational data, generated statistical outputs and summarized quantitative findings to support manuscript interpretation.

PU-Boost: Positive-Unlabeled Learning for Undiagnosed Hypertension
(First Author & Corresponding Author, Preprint in medRxiv) — *Lead Researcher, May 2026*
• Developed a two-stage framework combining Elastic Net and Random Forest for sample reconstruction, followed by XGBoost training on data from 43,105 adults (81.7% sensitivity; 41.1% fewer false negatives than standard XGBoost).
• Led overall study design, statistical model construction, model validation, result visualization and full-manuscript composition.

• Benchmarked PU-Boost against multiple baseline ML models including standard XGBoost and a neural network, evaluated screening-oriented classification metrics for clinical risk-stratification scenarios.
• Performed SHAP-based feature attribution analysis and interpreted meaningful predictors for hypertension risk prediction.

Digital Therapeutics in Dementia Care: A Mixed-Methods Study
(Fourth Author, Manuscript under review) *Data Analyst, Dec 2025*
• Analyzed adoption status of digital therapeutics (DTx) among 388 medical professionals from 196 cognitive care centers, applied ordinal logistic regression for survey data.
• Performed thematic analysis on 22 semi-structured interview transcripts, extracted insights on barriers and facilitators of digital-therapy implementation in dementia care.
• Integrated quantitative outputs and qualitative thematic findings, synthesized mixed-methods evidence to support manuscript discussion and conclusion sections.

Obesity Paradox in Anxiety and Depression
(First Author & Corresponding Author, Manuscript under review) *Lead Researcher, Aug 2025*
• Led study conceptualization, methodological design, model development, validation and code implementation for mental health epidemiological analysis.
• Performed sex-stratified multivariable logistic regression analyses on 43,099 adult participants, evaluated associations between multiple adiposity metrics and anxiety-depression mental-health outcomes.
• Modeled potential non-linear relationships by implementing restricted cubic spline functions, computed marginal predicted probabilities to quantify outcome risks across varying adiposity levels.
• Stratified statistical outputs by gender subgroups, interpreted adjusted odds-ratio results to examine sex-specific heterogeneity of the observed obesity-paradox pattern.

Factors Associated with patients' Healthcare-Seeking Behavior and Related Clinical Outcomes under China's Hierarchical Healthcare Delivery System
(Third Author, Published in Frontiers in Public Health) *Data Analyst, Apr 2024*
• Reviewed dataset of 9,098 healthcare encounters with cardiovascular risks, grounded in Andersen' s healthcare-utilization behavioral model.
• Analyzed multivariate logistic-regression outputs to examine socioeconomic, insurance and comorbidity predictors for patients' medical-institution-selection preferences.
• Compared blood-pressure, blood-glucose and lipid clinical-control outcomes between primary-care facilities and Grade-A hospitals for chronic-disease populations.

LANGUAGES
• Mandarin Chinese (Native), English (Professional working proficiency; TOEFL iBT: 102), Cantonese (Elementary)

PROFICIENT
• PADI Rescue Diver (2025) & AHA Heartsaver First Aid CPR AED (2025)

LEADERSHIP
• Captain, CUEB Ski Team; Vice President, Ice & Snow Sports Association; led team to 8th place in the 2025 Beijing Collegiate Alpine Slalom.

TECHNICAL SKILLS
• **Programming**: Python, R, SQL, Bash/Shell, Java, C
• **Statistical Methods**: Generalized Linear Models, Bayesian Modeling, Survival Analysis, Multivariate Analysis, Restricted Cubic Splines, Mixed-Effects Models
• **Machine Learning**: Elastic Net, Random Forest, XGBoost, Neural Networks, Positive-Unlabeled Learning
• **Bioinformatics**: RNA-seq, scRNA-seq, Differential Expression Analysis, GEO/NCBI Data Processing, GSEA, Cytoscape
• **Computing & Tools**: Linux, SLURM/HPC, Git/GitHub, scikit-learn, DESeq2, STAR, featureCounts, RStudio, VS Code, LaTeX, Quarto
• **Software**: Microsoft Excel, PowerPoint, Stata, EndNote